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Accelerating Power Flow Calculations in LV Networks Using Physics-Informed Graph Neural Networks

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***Abstract*—Precise calculations of network states and parameters are crucial for the modeling, monitoring, and operation of the distribution grid. AC power flow calculations are the preferred method for determining grid operating limits and assessing system hosting capacity. Traditional numerical power flow solvers are computationally intensive for large networks due to their iterative nature. In this context, we propose a physics-informed graph neural network technique as a fast alternative to the iterative power flow calculation methods. Our approach leverages the inherent physical knowledge of power systems through the integration of different physical constraints inside the model architecture and loss function. By incorporating physical laws such as Kirchhoff's law and active power balance as prior knowledge, the proposed method maintains high accuracy while significantly reducing computational time. The results show that the proposed model achieves better performance and higher generalizability than off-the-shelf data-driven approaches.**

***Index Terms*—Distribution grids, power flow analysis, physics-informed neural networks.**

I. Introduction

In power flow (PF) analysis, the primary objective is to determine the system's voltages and currents under specific load conditions. The distribution system operator needs to periodically analyze these network parameters to assess operating conditions and identify any problematic parts within the network. Traditional numerical power flow (PF) solvers, such as the Newton-Raphson method, are computationally expensive for large networks, with each minor change in grid configuration requiring a new calculation.

Deep learning (DL) approaches can provide faster PF solutions by learning intricate patterns from historical data [1]. However, these methods often struggle to capture the complex dynamics and constraints of electrical networks and typically require extensive training data. Most DL-based approaches proposed for solving the PF problem utilize models based on fully connected neural networks (FCNN) [2]. While FCNNs are simpler to implement, they have limited ability to adapt to changes in network topology. In contrast, graph neural networks (GNN) leverage the inherent structure of the network, making them adaptable to grids of varying sizes and configurations. A comprehensive review in [3] explores the application of GNNs in power systems, particularly highlighting their effectiveness in fault detection and PF analysis tasks. In [4], a Graph Attention Network is employed to perform the PF analysis. This helps in improving the learning process by assigning varying levels of importance to individual buses, thereby enabling the model to prioritize critical buses.

Most DL-based methods used in power systems are largely agnostic to the underlying physical principles. This oversight can lead to imprecise estimates, particularly when working with limited data or facing conditions that differ from those observed during the training phase. This underscores the need for advanced methods that combine DL-based approaches with domain-specific knowledge to improve prediction accuracy. In this context, physics-informed neural networks (PINN) approaches present a promising solution for solving complex problems in various scientific domains including the power system. PINNs leverage system equations to guide neural network learning, ensuring that predictions comply with underlying physical laws. Additionally, PINNs require less training data due to the integration of a physics-based regularization term, which effectively confines the solution space to physically plausible outcomes.

Previous research has investigated the use of PINNs in various power system tasks, including parameter estimation, dynamic state estimation, and fault detection [4], [5]. The authors in [6] propose a PINN approach that integrates the AC PF equation into the model's loss function to perform state estimation for partially observable low-voltage (LV) networks. However, this approach requires model retraining for new topologies which emphasizes the need for more flexible and adaptable solutions. Most of the current literature on PINN focuses on transmission networks, while their effectiveness in LV network problems remains largely unexplored. Therefore, it is essential to develop faster and more generalizable PINN-based methods specifically tailored for three-phase LV networks.

In this paper, we propose a novel power flow-informed graph neural network (PFiGNN) method to achieve a faster approximation of the PF problem by effectively determining voltages and currents in the LV network. Our objective is to maximize the benefits of data-based learning while ensuring compliance with the underlying physical laws.

The main contributions of this paper are:

- We developed a physics-informed graphical learning method that combines data-based and physics-based learning to efficiently and accurately perform PF calculations for three-phase LV networks.
- The proposed approach incorporates physical constraints into both the model's architecture and its loss function.
- The model can handle grids with varying topologies while maintaining high accuracy.

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利用物理信息图神经网络加速低压网络中的潮流计算

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摘要——精确计算网络状态和参数对于配电网的建模、监测和运行至关重要。交流潮流计算是确定电网运行极限和评估系统承载能力的首选方法。传统的数值潮流求解器由于其迭代特性，在大规模网络中计算量巨大。为此，我们提出了一种基于物理信息的图神经网络技术，作为迭代潮流计算方法的快速替代方案。我们的方法通过在模型架构和损耗函数中集成不同的物理约束，充分利用了电力系统固有的物理知识。通过将基尔霍夫定律和有功功率平衡等物理定律作为先验知识，所提出的方法在保持高精度的同时，显著降低了计算时间。结果表明，与现成的数据驱动方法相比，所提出的模型具有更好的性能和更高的泛化能力。

索引词——配电网、潮流分析、物理信息神经网络。

在潮流分析中，主要目标是确定特定负载条件下系统的电压和电流。配电系统运营商需要定期分析这些网络参数，以评估运行状况并识别网络中的任何问题部分。传统的数值潮流计算方法，例如牛顿-拉夫逊法，对于大型网络而言计算量巨大，电网配置的每一次微小变化都需要重新计算。

深度学习 (DL) 方法可以通过学习历史数据中的复杂模式来提供更快的功率因数 (PF) 解决方案 [1]。然而，这些方法通常难以捕捉电力网络的复杂动态特性和约束条件，并且通常需要大量的训练数据。大多数用于解决功率因数问题的基于深度学习的方法都采用基于全连接神经网络 (FCNN) 的模型 [2]。虽然 FCNN 实现起来更简单，但它们适应网络拓扑结构变化的能力有限。相比之下，图神经网络 (GNN) 利用网络的固有结构，使其能够适应不同规模和配置的电网。文献 [3] 中的一篇综述全面探讨了 GNN 在电力系统中的应用，尤其强调了其在故障检测和功率因数分析任务中的有效性。文献 [4] 采用图注意力网络进行功率因数分析。该网络通过为各个母线分配不同的重要性级别来改进学习过程，从而使模型能够优先处理关键母线。

大多数用于电力系统的深度学习方法在很大程度上忽略了其背后的物理原理。这种疏忽会导致估计结果不够精确，尤其是在数据有限或实际工况与训练阶段观测到的工况不同的情况下。这凸显了开发将深度学习方法与领域特定知识相结合的先进方法以提高预测精度的必要性。在此背景下，物理信息神经网络（PINN）方法为解决包括电力系统在内的各个科学领域的复杂问题提供了一种很有前景的解决方案。PINN 利用系统方程来指导神经网络的学习，确保预测结果符合物理定律。此外，由于引入了基于物理的正则化项，PINN 所需的训练数据更少，该正则化项有效地将解空间限制在物理上合理的范围内。

以往的研究已探讨了部分可观测网络（PINN）在电力系统各种任务中的应用，包括参数估计、动态状态估计和故障检测[4] [5]。文献[6]提出了一种 PINN 方法，该方法将交流功率因数方程集成到模型的损失函数中，用于对部分可观测低压（LV）网络进行状态估计。然而，该方法需要针对新的网络拓扑结构重新训练模型，这凸显了开发更灵活、更具适应性的解决方案的必要性。目前关于 PINN 的大部分文献都集中在输电网络，而其在低压网络问题中的有效性仍有待深入研究。因此，开发专门针对三相低压网络的、速度更快、更具通用性的 PINN 方法至关重要。

本文提出了一种新型的基于潮流信息的图神经网络（PFiGNN）方法，通过有效确定低压网络中的电压和电流，实现对潮流问题的快速逼近。我们的目标是在确保符合基本物理定律的前提下，最大限度地发挥数据驱动学习的优势。

本文的主要贡献有：

- 我们开发了一种基于物理的图形学习方法，该方法结合了基于数据的学习和基于物理的学习，可以高效、准确地对三相低压网络进行功率因数计算。
- 所提出的方法将物理约束纳入模型的架构及其损失函数中。
- 该模型能够处理具有不同拓扑结构的网格，同时保持高精度。

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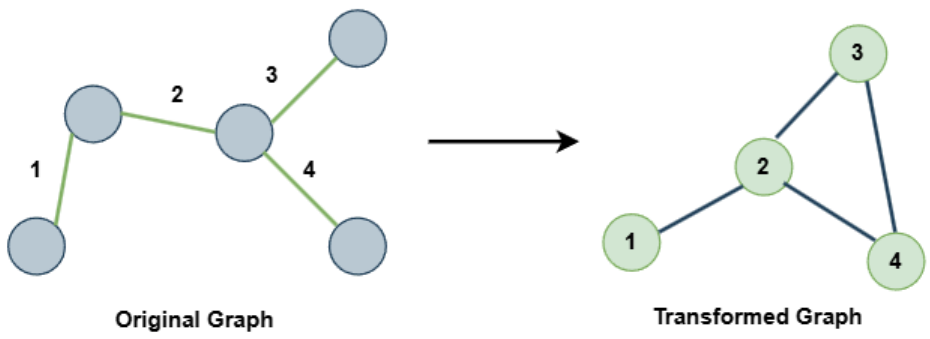


Fig. 1: Simple illustration of the transformation from bus graph to line graph representation.

The paper is structured as follows: Section II provides a description of the proposed methodology employed in this study. Experimental results are discussed in detail in Section III, followed by a conclusion in Section IV.

II. Methodology

This section discusses how the power flow formulation is interpreted within the context of GNN algorithms and presents the proposed PFiGNN model.

A. Interpreting Power Flow as a GNN Problem

Electrical networks are inherently graph-like in their structure which makes GNN a suitable choice for their modeling. The network topology can be represented as an undirected graph $G(N, E)$, where the set of graph nodes N corresponds to network buses and the set of edges E represents power lines. The node features encompass various bus attributes, including voltage magnitude and active and reactive power injections. Similarly, the edge features contain the resistance and reactance values for each line. In GNN, message passing [7] serves as a communication protocol. It allows nodes to iteratively update their features by aggregating information from neighboring nodes. Intuitively, the fundamentals of PF in LV networks naturally align with the message passing strategy employed by GNN. Given active $\boldsymbol{P}_i$ and reactive $\boldsymbol{Q}_i$ power injections linked to bus i of the network, along with admittance matrix $\boldsymbol{Y}$ as the input features, our objective is to determine the output bus voltages $\boldsymbol{V}_i$ and line currents $\boldsymbol{I}_{ij}$, thereby transforming the PF problem into a node/edge regression task.

In this work, we propose an alternative dual representation of network graph. This representation leverages both the original network graph and a transformed line graph generated by a new *graph transformation layer*. Fig. 1 shows the transition from the original bus graph to its corresponding line graph. In the original graph, buses are connected by electrical lines, whereas in the transformed graph, lines become the vertices and buses form the edges. This dual graph representation, inspired by the idea presented in [8], offers a more comprehensive view of the network and enables targeted optimization for both bus connectivity and line configurations.

B. Proposed Power Flow-informed GNN Model

The proposed model, termed PFiGNN, is based on integrating the physics of LV networks into the GNN algorithm. The

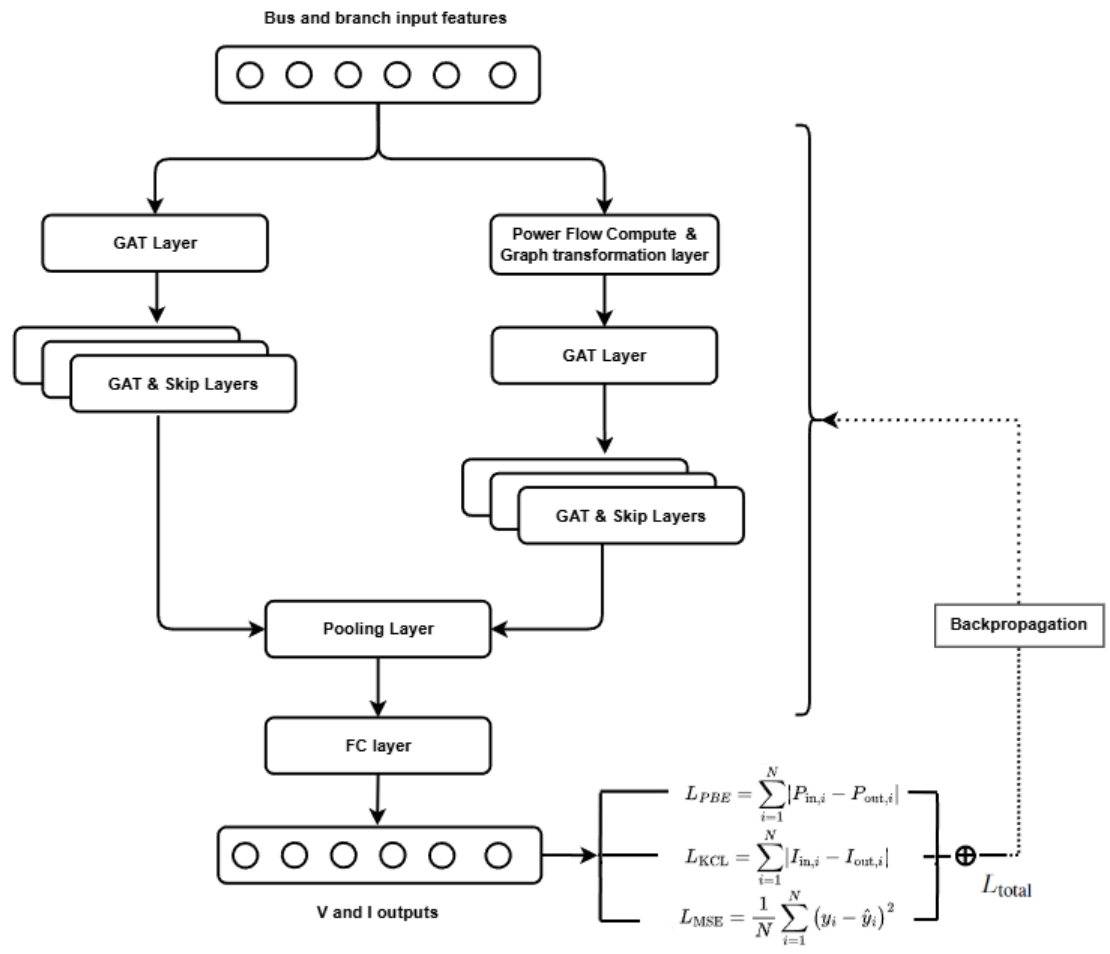


Fig. 2: Proposed PFiGNN Model

architecture includes a *Graph Attention Network* (GAT) layer, a specialized power flow computation layer and a graph transformation layer. Input features $\boldsymbol{X}$ including power injections and line impedances are processed through two parallel paths as shown in Fig. 2. Initially, the GAT layer processes these input features $\boldsymbol{X}$ and produces a new set of node features (or embeddings) $\boldsymbol{X'}$ using the attention mechanism a. This mechanism enables each node to rank its neighboring nodes in the graph by assigning them different attention coefficients that reflect their individual relevance. The attention coefficients α_{ij} for node i with respect to its neighboring node j are given as:

$$e_{ij} = a(\mathbf{w}\boldsymbol{X}_i, \mathbf{w}\boldsymbol{X}_j), \quad \alpha_{ij} = \text{softmax}_j(e_{ij}) \tag{1}$$

where $\mathbf{w}$ represents the learnable weight matrix that transforms input feature vectors into a new feature space. The term e_{ij} represents the raw attention score, which is subsequently normalized using the softmax function to yield the attention coefficients α_{ij}. These coefficients quantify the influence of neighboring node j on node i. The final output features of node i through the GAT layer are given as:

$$\boldsymbol{X}'_i = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \boldsymbol{X}_j\right) \tag{2}$$

where σ denotes the $ReLU$ activation function, which introduces non-linearity into the model to capture complex patterns. The GAT layer enables the proposed model to focus on the most relevant aspects of the graph to produce accurate results.

The *Power Flow Computation Layer* in the model architecture computes the approximate active and reactive power

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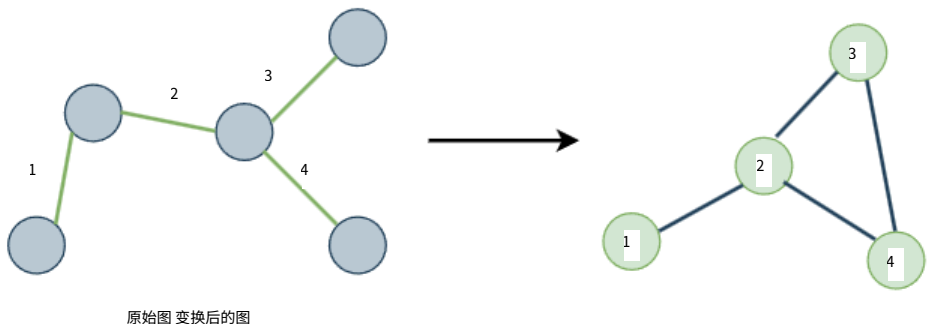


图 1：从公交图到线图表示的转换的简单示例。

本文结构如下：第二节描述了本研究采用的方法；第三节详细讨论了实验结果；第四节给出了结论。

二、M

本节讨论如何在 GNN 算法的背景下解释潮流公式，并提出所提出的 PFiGNN 模型。

A. 将潮流问题解释为 GNN 问题

电力网络本质上具有图状结构，这使得图神经网络（GNN）成为对其进行建模的理想选择。网络拓扑可以表示为一个无向图 G(N, E)，其中图的节点集合 N 对应于网络母线，边集合 E 对应于电力线。节点特征包含各种母线属性，例如电压幅值以及有功功率和无功功率注入。类似地，边特征包含每条线路的电阻和电抗值。在 GNN 中，消息传递[7]用作通信协议。它允许节点通过聚合来自相邻节点的信息来迭代更新其特征。直观地说，低压网络中的功率因数（PF）基本原理与 GNN 采用的消息传递策略自然契合。给定与网络母线 i 相连的有功功率 P 和无功功率 Q，以及导纳矩阵 Y 作为输入特征，我们的目标是确定输出母线电压 V 和线路电流 I，从而将功率因数问题转化为节点/边回归任务。

本文提出了一种网络图的替代对偶表示方法。该表示方法同时利用了原始网络图和由新的图变换层生成的变换线图。图 1 展示了从原始母线图到其对应线图的转换过程。在原始图中，母线通过电线连接；而在变换后的图中，电线成为顶点，母线则构成边。这种对偶图表示方法借鉴了文献[8]中的思想，能够更全面地展现网络结构，并支持针对母线连通性和线路配置进行针对性优化。

B. 提出的基于潮流信息的图神经网络模型

所提出的模型，称为 PFiGNN，是基于将 LV 网络的物理特性融入 GNN 算法而构建的。

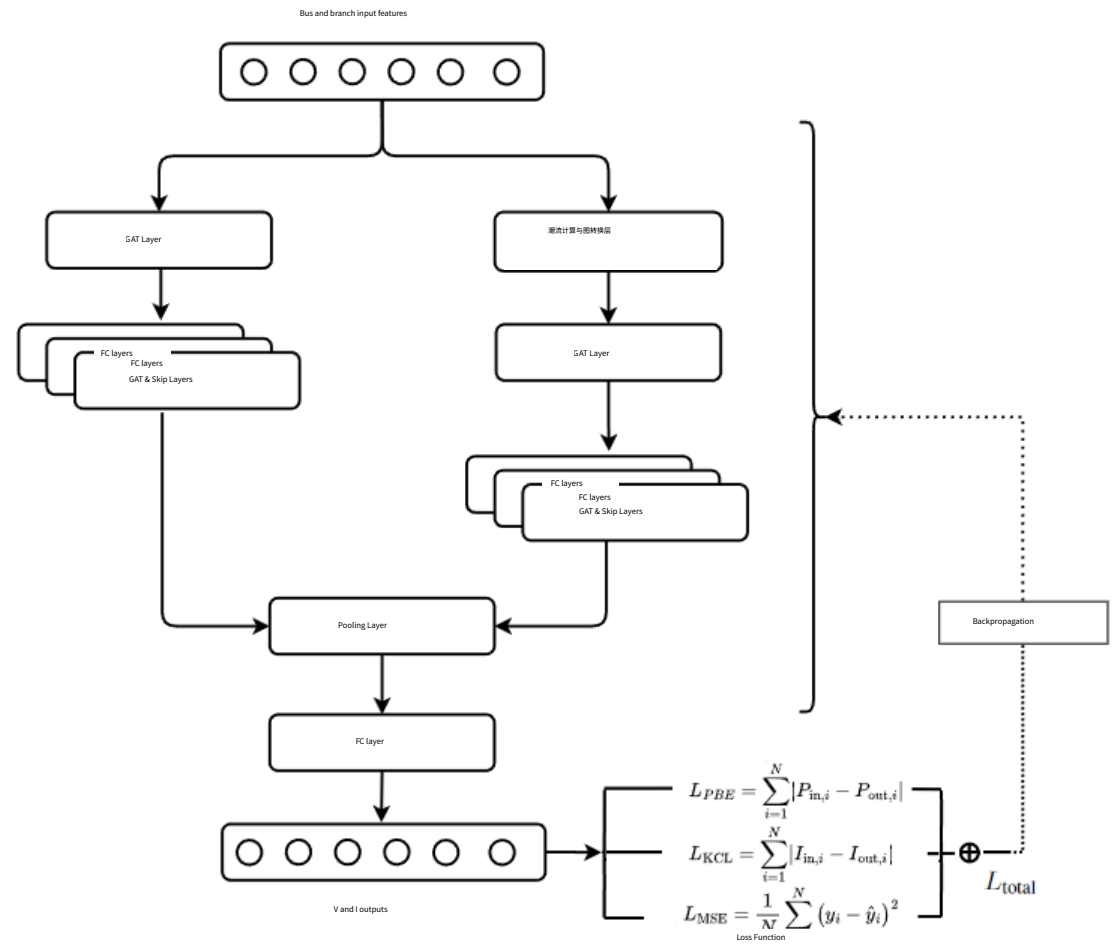


图 2：提出的 PFiGNN 模型

该架构包含一个图注意力网络（GAT）层、一个专用潮流计算层和一个图变换层。输入特征 X（包括功率注入和线路阻抗）通过两条并行路径进行处理，如图 2 所示。首先，GAT 层处理这些输入特征 X，并使用注意力机制 a 生成一组新的节点特征（或嵌入）X。该机制使每个节点能够通过为其邻居节点分配不同的注意力系数来对图中的邻居节点进行排序，这些系数反映了它们各自的相关性。节点 i 相对于其邻居节点 j 的注意力系数 α 如下所示：

$$e = a(wX, wX), \alpha = \text{softmax}(e) \quad (1)$$

其中 w 表示可学习的权重矩阵，它将输入特征向量转换到新的特征空间。er 表示原始注意力得分，随后使用 softmax 函数对其进行归一化，得到注意力系数 α。这些系数量化了相邻节点 j 对节点 i 的影响。节点 i 通过 GAT 层的最终输出特征如下所示：

$$X = \sigma \left(\boxtimes \underset{j \in N(i)}{X} \text{斧头} \boxtimes \right) \quad (2)$$

其中 σ 表示 ReLU 激活函数，它为模型引入非线性以捕捉复杂模式。GAT 层使所提出的模型能够专注于图中最相关的方面，从而产生准确的结果。

模型架构中的潮流计算层
结构计算近似的有功功率和无功功率

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PQ-flows across the edges of the grid graph. This computation leverages the mathematical formulations of nodal power balance given in [9] through the following equations:

$$\sum_{k:j\to k} P_{jk} \simeq P_{ij} + p_j \tag{3}$$

$$\sum_{k:j\to k} Q_{jk} \simeq Q_{ij} + q_j \tag{4}$$

The equations 3 and 4 state that the sum of all power flows from node j to its connected nodes k is approximately equal to the power flowing into node j from node i plus the power injected at node j. To compute the line PQ-flows based on these equations, we utilize a backward sweep methodology where we accumulate power demands from terminal nodes towards their parent nodes. The power flow computation layer systematically allocates these line PQ-flows to the corresponding edges of the original graph. This enables the direct incorporation of the fundamental principle of PF equilibrium into the model architecture. Next, the graph transformation layer assigns the PQ-flows to the vertices of the new line graph and passes them as input to the subsequent GAT layer. This allows the right branch of the model to use the line graph representation, while the left branch uses the original graph.

GAT layers primarily focus on learning node-specific features. To capture global features from the entire graph, we incorporate a mean pooling layer that aggregates information from the final GAT layers across both model branches. This allows the model to learn both local (i.e. node-level) and global (i.e. network-level) features. To mitigate the oversmoothing issue in the GNNs [8], where node representations become increasingly similar as model depth increases, we employ skip connections. These connections allow information from earlier layers to flow directly to deeper layers which helps in preserving unique features for each node. A fully connected (FC) layer is added to get the final voltage and current outputs.

C. Physics-informed Loss Function

At the core of our PFiGNN model lies the implementation of a physics-informed loss function. We incorporated different physical constraints into the model loss function which guides it to produce physically plausible solutions. In this study, we implemented constraints based on Kirchhoff's Current Law (KCL) and nodal power balance into the model's loss function using algebraic equations as illustrated in Fig. 2. During training, the model is optimized to minimize both the discrepancy between predicted and true values, as well as the violation of the imposed physics-based constraints. This optimization is achieved through backpropagation, a technique that iteratively adjusts the model parameters to minimize the loss function. The overall loss function is expressed as:

$$L_{\text{total}} = L_{\text{MSE}} + \alpha \cdot L_{\text{KCL}} + \beta \cdot L_{\text{PBE}} \tag{5}$$

where L_{MSE} represents the data-driven part of the loss function. The physical term L_{KCL} quantifies the KCL constraint adherence, which validates that the sum of currents entering and leaving a node is zero. The second physical term L_{PBE} represents the total active power balance error, calculated as the sum of absolute differences between incoming and outgoing active power at each node in the network. The active power is derived from the predicted voltages and currents, i.e. $P = \hat{V}\hat{I}\cos(\theta)$. To balance the contribution of physical loss terms during training, loss weights α and β are introduced. This prevents any single physical loss term from dominating the overall training process. By including these physical constraints in the objective function, the model's learning process is guided toward solutions that are consistent with the characteristics of electrical networks.

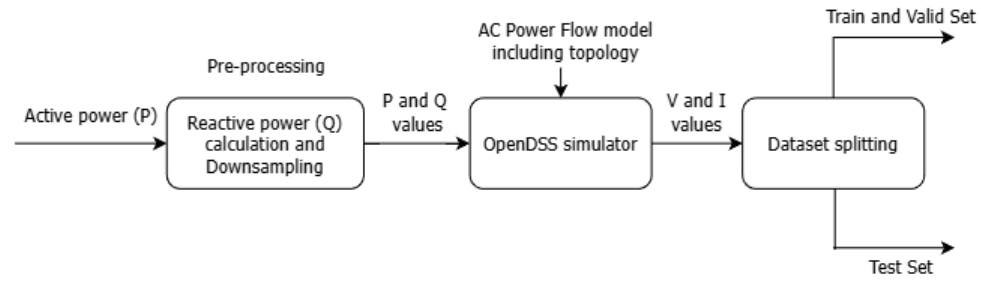


Fig. 3: Overview of the data preparation scheme

III. Experimental Setup and Results

We evaluated the proposed model on a non-synthetic European-style LV grid documented in [10]. For our analysis, we selected three feeders of varying complexity from the grid dataset: the 14-bus (`case14`), 16-bus (`case16`) and 18-bus (`case18`) configuration. Fig. 4 illustrate the layouts of the 14-bus and 18-bus feeders. Active power load data is collected from the LV network of Flanders, Belgium over a period of 30 days and subsequently downsampled to 30-minute resolution. Reactive power profiles are generated by assuming a random power factor within the range of 0.95 to 0.98. Using downsampled power injection values and known network characteristics, we obtained target voltage and current values by simulating PF in OpenDSS software [11]. Fig. 3 illustrates the data preparation pipeline for model training. The dataset consisted of a total of $30\times48 = 1440$ PF snapshots. We divided it into three distinct subsets: 60% for training, 20% for validation and 20% for testing, to facilitate the model training and evaluation process. Moreover, the dataset was standardized using per unit (p.u.) normalization using base apparent power and voltage levels. Hyper-parameter tuning is beyond the scope of this study; therefore, we used standard parameters from the literature, including the *Adam* optimizer [12] with a learning rate of 0.001. We implemented the model using the PyTorch and PyTorch Geometric deep learning libraries.

A. Numerical Results

We compared the performance of the proposed PFiGNN model with other DL approaches under various conditions. The models used for comparison include standard FCNN and standard GNN, both of which are physics-agnostic. Furthermore, we introduced a physics-informed variant of the FCNN model,

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P Q 流经网格图的边缘。该计算利用了文献 [9] 中给出的节点功率平衡的数学公式，具体公式如下：

$$P \simeq P + \sum_{k:j\to k} p \tag{3}$$

$$Q \simeq Q + \sum_{k:j\to k} q \tag{4}$$

公式 3 和 4 表明，从节点 j 到其连接节点 k 的所有功率流之和近似等于从节点 i 流入节点 j 的功率加上在节点 j 注入的功率。为了基于这些公式计算线路功率流（PQ），我们采用反向扫描方法，将终端节点到其父节点的功率需求累加起来。功率流计算层将这些线路功率流（PQ）系统地分配到原始图的相应边上。这使得功率流平衡的基本原理能够直接融入模型架构中。接下来，图转换层将功率流（PQ）分配到新线路图的顶点，并将其作为输入传递给后续的图转换层（GAT）。这使得模型的右分支可以使用线路图表示，而左分支则使用原始图。

GAT 层主要侧重于学习节点特定的特征。为了从整个图中捕获全局特征，我们引入了一个均值池化层，用于聚合来自模型两个分支的最终 GAT 层的信息。这使得模型能够学习局部（即节点级）和全局（即网络级）特征。为了缓解图神经网络（GNN）中存在的过度平滑问题[8]（即随着模型深度的增加，节点表示变得越来越相似），我们采用了跳跃连接。这些连接允许来自早期层的信息直接流向更深的层，从而有助于保留每个节点的独特特征。最后，我们添加了一个全连接（FC）层来获得最终的电压和电流输出。

C. 基于物理学的损失函数

PFiGNN 模型的核心在于实现了基于物理原理的损失函数。我们将不同的物理约束融入模型损失函数中，引导其生成符合物理规律的解。在本研究中，我们利用代数方程（如图 2 所示）将基于基尔霍夫电流定律（KCL）和节点功率平衡的约束条件融入模型损失函数中。在训练过程中，模型会进行优化，以最小化预测值与真实值之间的差异，以及对所施加的基于物理原理的约束条件的违反程度。这种优化是通过反向传播实现的，反向传播是一种迭代调整模型参数以最小化损失函数的技术。总损失函数表示为：

$$L = L + \alpha \cdot L + \beta \cdot L \tag{5}$$

其中 L 代表损失函数的数据驱动部分。物理项 L 量化了基尔霍夫电流定律 (KCL) 约束的满足程度，验证了流入电流之和满足 KCL 约束。

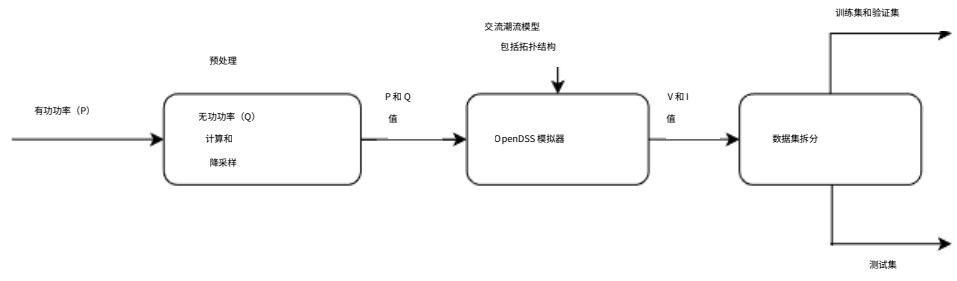


图 3：数据准备方案概述

离开节点的功率为零。第二个物理项 L 表示总有功功率平衡误差，计算方法为网络中每个节点输入和输出有功功率绝对差值之和。有功功率由预测的电压和电流导出，即 P = ˆV ˆI cos(θ)。为了平衡训练过程中各个物理损耗项的贡献，引入了损耗权重 α 和 β。这可以防止任何单一的物理损耗项主导整个训练过程。通过将这些物理约束纳入目标函数，模型的学习过程被引导至与电网特性相符的解。

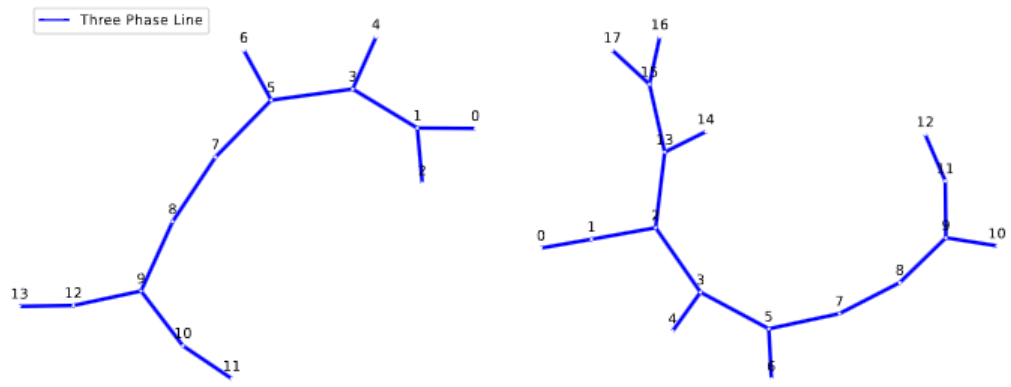
III. 电子自旋共振

我们在文献[10]中记录的非合成欧式低压电网上评估了所提出的模型。为了进行分析，我们从电网数据集中选取了三种不同复杂程度的馈线：14 节点（case14）、16 节点（case16）和 18 节点（case18）配置。图 4 展示了 14 节点和 18 节点馈线的布局。有功功率负荷数据采集自比利时弗兰德斯地区的低压电网，采集周期为 30 天，随后下采样至 30 分钟分辨率。无功功率曲线的生成基于 0.95 至 0.98 范围内的随机功率因数假设。利用下采样后的功率注入值和已知的网络特性，我们通过在 OpenDSS 软件[11]中模拟功率因数（PF）获得了目标电压和电流值。图 3 展示了模型训练的数据准备流程。数据集共包含 30×48=1440 个功率因数快照。为了便于模型训练和评估，我们将数据集分为三个不同的子集：60% 用于训练，20% 用于验证，20% 用于测试。此外，我们使用基准视在功率和电压水平，通过标幺值 (pu) 归一化对数据集进行了标准化。超参数调优超出了本研究的范围；因此，我们使用了文献中的标准参数，包括学习率为 0.001 的 Adam 优化器[12]。我们使用 PyTorch 和 PyTorch Geometric 深度学习库实现了该模型。

A. 数值结果

我们在各种条件下将所提出的 PFiGNN 模型与其他深度学习方法进行了性能比较。用于比较的模型包括标准 FCNN 和标准 GNN，它们都与物理无关。此外，我们还引入了一种物理信息化的 FCNN 模型变体。

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(a) `case14` (b) `case18`

Fig. 4: Illustration of 14-bus and 18-bus test feeders layout.

TABLE I: Comparing Different Models based on Denormalized MSE Scores for Phase-I Voltages and Currents.

Method	Fixed Grid Topology		Multiple Grid Topologies	
	Voltage	**Current**	**Voltage**	**Current**
Standard FCNN	0.151	1.751	0.961	4.160
Standard GNN	0.480	2.653	0.771	3.622
PIFCNN	**0.140**	**1.675**	0.928	3.934
Proposed PFiGNN	0.492	2.528	**0.602**	**3.358**

termed PIFCNN, by replacing the conventional loss function with a physics-informed loss function. This adjustment was made to evaluate how incorporating physical constraints in FCNN models affects its performance. We evaluated model accuracy in scenarios where the grid topology was fixed and in scenarios where the models were applied across multiple grids with different topologies. Table I displays the denormalized mean squared error (MSE) values for voltage and current predictions across both experimental scenarios.

1) Fixed Grid Topology: In the first experiment, we tested the proposed models on the same grid sample on which it was trained. For this task, samples of the `case18` grid are considered. This is a less challenging case in which the models only need to adapt to variation in values of a specific set of buses and lines, but not to changes in the underlying network topology. Table I shows that in a fixed topology scenario, the physics-informed FCNN model outperformed the proposed PFiGNN model, achieving approximately 50% lower voltage errors and 29% lower current errors. The advantage of the FCNN-based models lies in their ability to consistently learn fixed patterns from constant grid topologies, allowing them to optimize their weights more efficiently than the GNN models. Additionally, the added complexity of the proposed model increases the risk of overfitting in fixed topology scenarios. The boxplot in Fig. 6 displays the distribution of voltage values obtained by the proposed model for the `case18` grid.

2) Generalization Across Multiple Grid Topologies: The proposed model is capable of generalizing across grids of varying sizes and topologies. This allows a model trained on a specific grid topology to be effectively applied to new grids, regardless of their configuration or size. In the second experiment, the models are trained on samples from all three

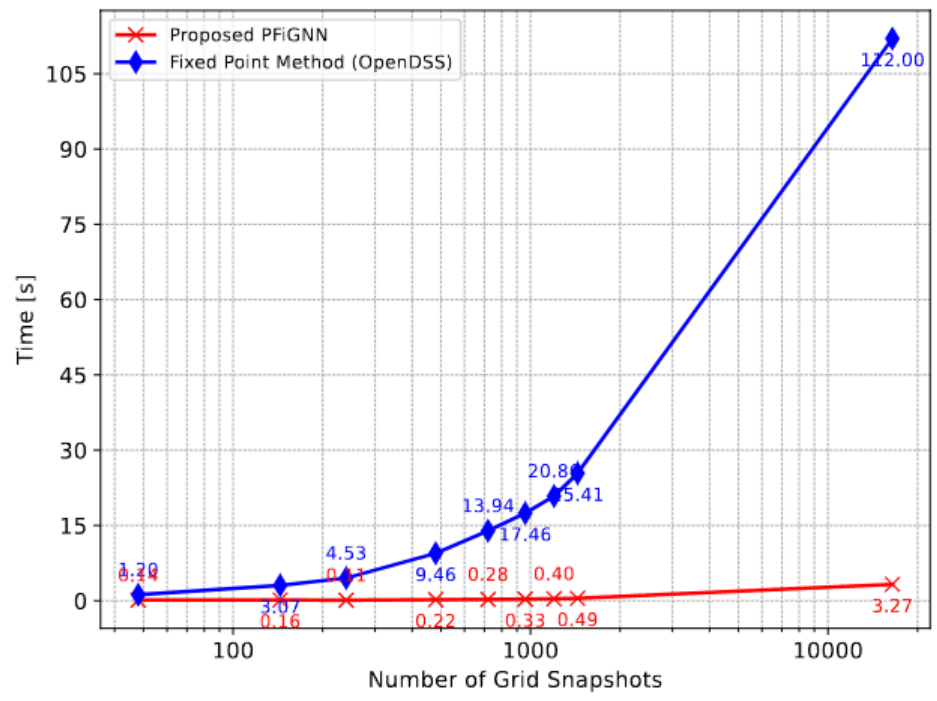


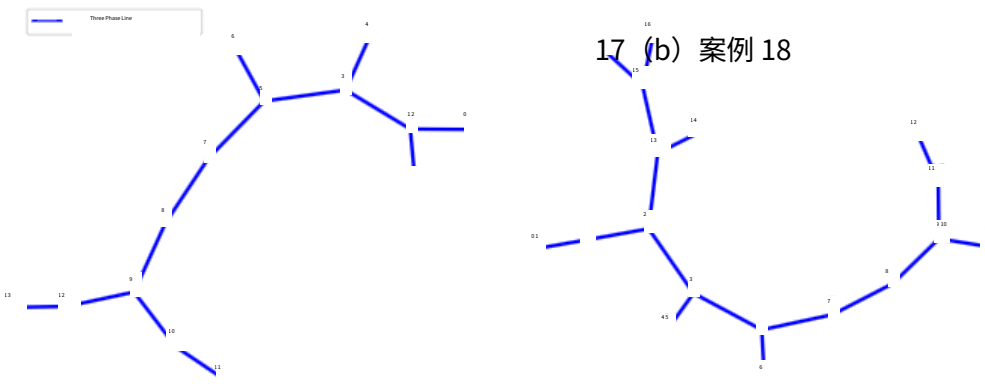
Fig. 5: Comparison of model evaluation times

feeders: `case14`, `case16`, and `case18` to evaluate their generalization performance. Unlike GNNs, the FCNN model cannot directly handle input data with varying dimensions or shapes. Therefore, to enable processing by the FCNN model, the input data is zero-padded to match the dimensions of the largest grid sample in the dataset. When evaluated on samples from three distinct feeders, the proposed PFiGNN model outperformed the FCNN model, achieving 35% lower voltage and 19% lower current errors. The physical loss function enables the proposed model to closely adhere to fundamental power system principles which results in more generalizable and reliable predictions across various network topologies. However, in this experiment, we observed slightly higher errors overall, particularly in current predictions. This can be attributed to the unique configurations and characteristics of the feeders, which introduce additional modeling complexity.

B. Time Complexity

The main advantage of the proposed method over traditional numerical solvers is its reduced time complexity, which is crucial for performing large-scale grid simulations. We conducted a comparative evaluation of the computational times between the proposed model and the fixed point current injection method. The fixed point method, an iterative approach for solving nonlinear equations, is effective in PF analysis due to its efficiency and reliable convergence. For comparison purposes, we evaluated multiple grid snapshots with varying load demands using `case14` grid. While the fixed point method's computation time increases linearly with grid snapshots, the proposed model's evaluation time remains nearly constant, as shown in Fig. 5. This distinction highlights our model's computational efficiency by removing the need for iterative convergence processes typical to numerical solvers. Notably, the final point in the plot shows the proposed model computing $16,464$ grid snapshots in a few seconds, effectively

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（a）案例 14

17（b）案例 18

图 4：14 母线和 18 母线测试馈线布局示意图。

表一：基于第一阶段电压和电流的非归一化 MSE 得分比较不同模型。

方法	固定网格拓扑结构		多网格拓扑结构	
	电压电流电压		当前的	
标准 FCNN	0.151 1.751		0.961	4.160
标准 GNN	0.480 2.653		0.771	3.622
PIFCNN	0.140 1.675		0.928	3.934
建议的 PFiGNN 0.492 2.528			0.602	3.358

通过将传统的损失函数替换为基于物理信息的损失函数，我们提出了一种称为 PIFCNN 的模型。此调整旨在评估在 FCNN 模型中加入物理约束对其性能的影响。我们评估了两种场景下的模型精度：一种是电网拓扑结构固定，另一种是模型应用于具有不同拓扑结构的多个电网。表 I 显示了两种实验场景下电压和电流预测的非归一化均方误差 (MSE) 值。

1) 固定电网拓扑：在第一个实验中，我们使用与训练模型相同的电网样本测试了所提出的模型。本实验采用案例 18 电网的样本。这是一个相对简单的案例，模型只需适应特定母线和线路数值的变化，而无需适应底层网络拓扑结构的变化。表 1 显示，在固定拓扑场景下，基于物理信息的 FCNN 模型优于所提出的 PFiGNN 模型，电压误差降低了约 50%，电流误差降低了约 29%。基于 FCNN 的模型的优势在于它们能够从恒定的电网拓扑结构中持续学习固定模式，从而比 GNN 模型更有效地优化权重。此外，所提出模型的复杂性增加，也增加了在固定拓扑场景下过拟合的风险。图 6 中的箱线图展示了所提出的模型在案例 18 电网中获得的电压值分布。

2）跨多种网格拓扑的泛化：

所提出的模型能够泛化到不同大小和拓扑结构的网格。这使得在特定网格拓扑结构上训练的模型可以有效地应用于新的网格，无论其配置或大小如何。在第二个实验中，模型使用来自所有三个网格的样本进行训练。

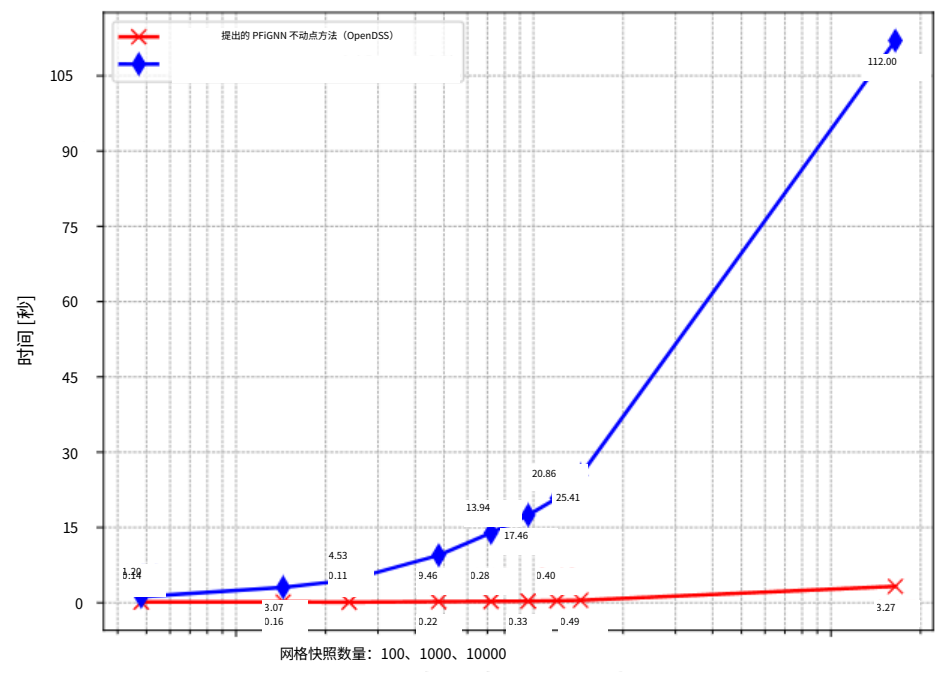


图 5：模型评估时间比较

为了评估其泛化性能，我们选取了馈线案例 14、案例 16 和案例 18。与 GNN 不同，FCNN 模型无法直接处理维度或形状各异的输入数据。因此，为了使 FCNN 模型能够处理这些数据，我们对输入数据进行了零填充，使其维度与数据集中最大电网样本的维度相匹配。在三个不同馈线的样本上进行评估时，我们提出的 PFiGNN 模型优于 FCNN 模型，电压误差降低了 35%，电流误差降低了 19%。物理损耗函数使得所提出的模型能够紧密遵循电力系统的基本原理，从而在各种网络拓扑结构中实现更具泛化性和可靠性的预测。然而，在本实验中，我们观察到总体误差略高，尤其是在电流预测方面。这可能是由于馈线的独特配置和特性引入了额外的建模复杂性。

B. 时间复杂度

与传统数值求解器相比，所提出方法的主要优势在于其更低的时间复杂度，这对于执行大规模网格模拟至关重要。我们对所提出的模型和定点电流注入法的计算时间进行了比较评估。定点法是一种求解非线性方程的迭代方法，由于其高效性和可靠的收敛性，在 PF 分析中非常有效。为了进行比较，我们使用 case14 网格评估了具有不同负载需求的多个网格快照。如图 5 所示，虽然定点法的计算时间随网格快照数线性增加，但所提出的模型的评估时间几乎保持不变。这一差异突显了我们模型的计算效率，因为它无需数值求解器典型的迭代收敛过程。值得注意的是，图中的最后一个点显示，所提出的模型在几秒钟内计算了 16464 个网格快照，效率极高。

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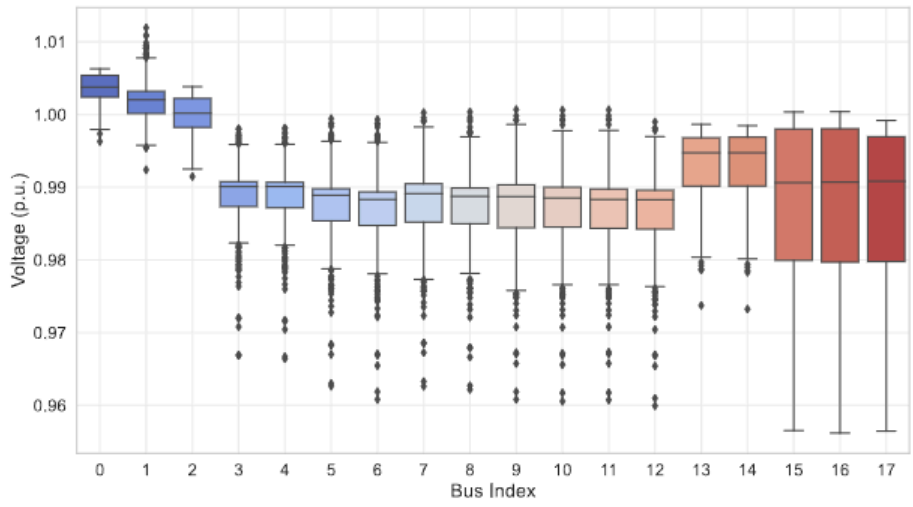


Fig. 6: Boxplot showing the bus voltage predictions for `case18` branched feeder. Voltage decreases with increasing distance from the source node due to a higher voltage drop.

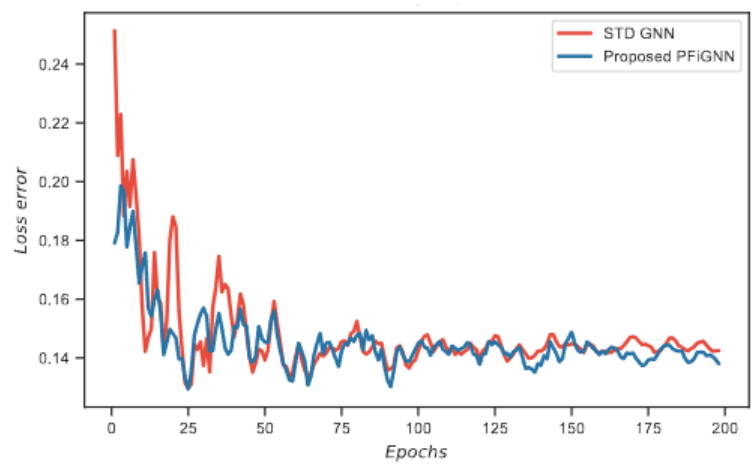
simulating nearly a full year of PF estimates. The comparison was performed on an Intel Core i7 (16 CPU cores) machine.

C. Assessment of Model Physical Consistency

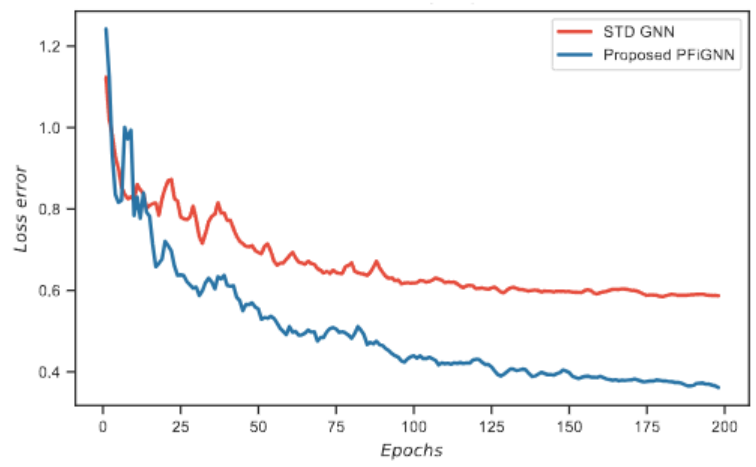
The purpose of using a PINN-based model is not only to minimize voltage and current prediction error but to also learn physical constraints as an auxiliary objective. Therefore, it is essential to evaluate the model's performance across both data-driven and physics-based learning tasks. To assess the impact of the physics-informed loss terms on the learning process, we compared the validation loss curves of the standard GNN and the proposed PFiGNN models, as shown in Fig. 7. In the physics-agnostic standard GNN model, network training relies exclusively on the data-driven MSE loss term, with the additional physical loss term (i.e. KCL error) calculated solely for monitoring purposes. Conversely, the PFiGNN model incorporates both the MSE and KCL loss terms during the training process. As illustrated in Fig. 7a, the validation MSE loss of the proposed PFiGNN model shows a slight improvement over the standard GNN model. However, the proposed model significantly outperforms the physics-agnostic GNN model in learning the KCL constraint, as shown in Fig. 7b. Consequently, in out-of-sample test scenarios, the standard GNN model is more likely to violate fundamental physical principles, such as KCL, leading to physically inconsistent predictions and reduced generalization performance.

IV. Conclusion

We presented a novel PINN-based approach for faster power flow calculations in the LV network by embedding fundamental power system principles into the GNN model. The incorporation of physical constraints acts as a regularization mechanism, reducing the risk of overfitting and improving the model's ability to generalize across a range of network topologies. Results showed that our model outperformed benchmark DL-based approaches in both accuracy and convergence. It also demonstrates superior computational efficiency compared to traditional numerical approaches. In future work, we plan to evaluate the model in limited observability scenarios and perform hyperparameter tuning to improve its performance.



(a) MSE Error



(b) KCL Error

Fig. 7: Comparison between the loss components of physics-aware and physics-blind models.

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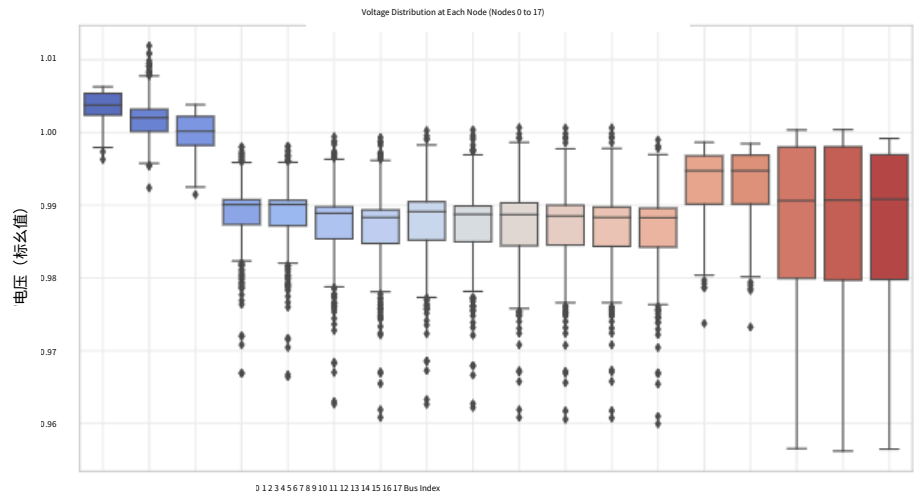
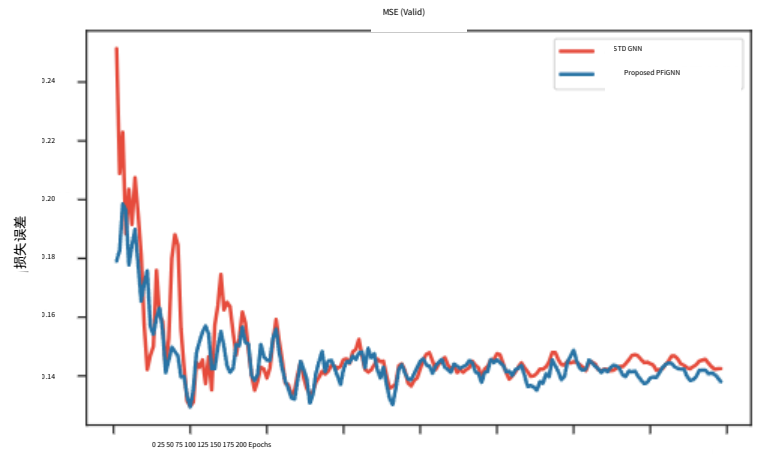


图 6：箱线图显示了案例 18 分支馈线的母线电压预测结果。由于电压降增大，电压随着距源节点距离的增加而降低。

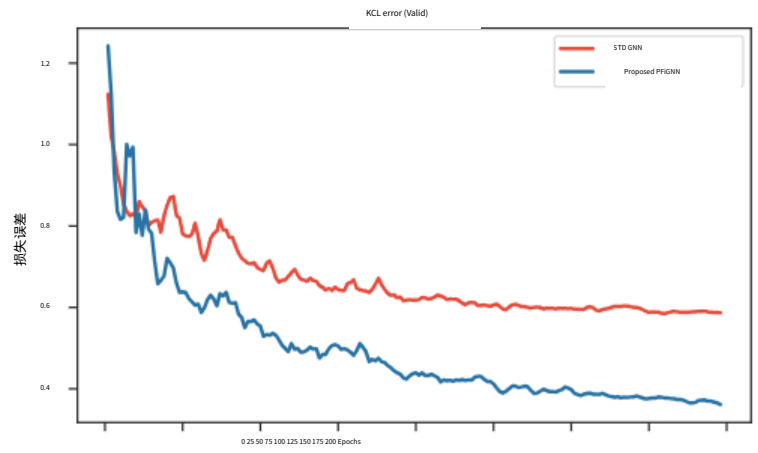
模拟了近一整年的 PF 估算值。该对比测试是在一台配备英特尔酷睿 i7 处理器（16 个 CPU 核心）的机器上进行的。

C. 模型物理一致性评估

使用基于 PINN 的模型的目的不仅在于最小化电压和电流预测误差，还在于学习物理约束作为辅助目标。因此，评估模型在数据驱动和基于物理的学习任务中的性能至关重要。为了评估物理信息损失项对学习过程的影响，我们比较了标准 GNN 模型和所提出的 PFiGNN 模型的验证损失曲线，如图 7 所示。在不考虑物理因素的标准 GNN 模型中，网络训练完全依赖于数据驱动的 MSE 损失项，而额外的物理损失项（即 KCL 误差）仅用于监测目的。相反，PFiGNN 模型在训练过程中同时考虑了 MSE 和 KCL 损失项。如图 7a 所示，所提出的 PFiGNN 模型的验证 MSE 损失相比标准 GNN 模型略有改进。然而，如图 7b 所示，所提出的模型在学习 KCL 约束方面显著优于不考虑物理因素的 GNN 模型。因此，在样本外测试场景中，标准 GNN 模型更有可能违反基本物理原理，例如基尔霍夫电流定律 (KCL)，从而导致物理上不一致的预测和泛化性能降低。



（a）均方误差



（b）基尔霍夫电流定律误差

图 7：物理感知模型和物理盲模型的损失组成部分比较。

IV. C

我们提出了一种基于 PINN 的新型低压电网潮流计算方法，该方法通过将基本电力系统原理嵌入 GNN 模型中来实现。物理约束的引入起到了正则化的作用，降低了过拟合的风险，并提高了模型对各种网络拓扑结构的泛化能力。结果表明，我们的模型在精度和收敛性方面均优于基准深度学习方法。此外，与传统数值方法相比，我们的模型还展现出更高的计算效率。在未来的工作中，我们计划在可观测性受限的场景下评估该模型，并进行超参数调优以进一步提升其性能。

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